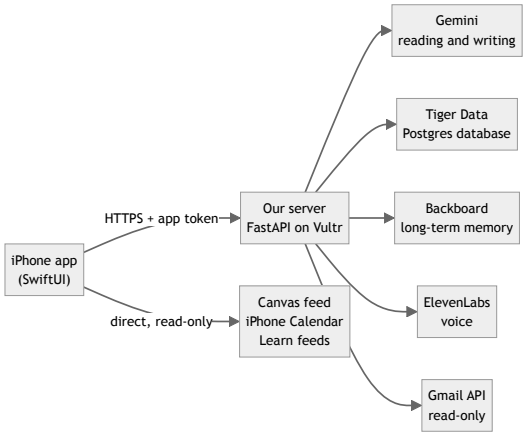
HABITS · A STUDENT STARTER PACK

Habits under the hood

Which APIs the app talks to, and what runs on the phone and the server without you tapping anything.

How it fits together

The iPhone app never holds an AI or database key. It talks to one server of ours, which calls everything else.



Outside services

SERVICE	WHAT IT DOES HERE	CALLED BY
Gemini	Turns messy text, voice, photos (flyers, receipts) and email into structured data; writes summaries, the briefing and chat answers. Tries six models in turn, 25 s each, when one is busy or out of quota.	Server
Tiger Data	The database: cached summaries, voice logs and spending (time-series hypertables), the weekly budget, and the history Ask answers "when did I…" from.	Server
Backboard	Long-term memory about you ("interested in robotics"). Searched before chat, recall and Capture; editable in the Memory Inspector.	Server
ElevenLabs	Speaks the daily briefing and chat replies (`eleven_multilingual_v2`).	Server
Gmail API	Reads the last two days of inbox mail, including Teams notifications. Read-only scope: it can't send, delete or change mail.	Server
Vultr	Hosts the server: Ubuntu, Caddy for HTTPS, systemd keeps it running.	—
Canvas	Your course calendar feed (.ics link), or the Canvas API if you add a token.	Phone
iPhone Calendar	Events from your calendars (incl. school Outlook) through Apple's EventKit, on the phone.	Phone
Learn feeds	arXiv, Hacker News (Algolia search), Hugging Face daily papers and blog RSS for the Learn tab.	Phone
Apple Speech	Mic dictation, on-device when the phone supports it. Audio never leaves the phone.	Phone

Server endpoints

Every call needs the app's bearer token. Base address: `https://45-76-234-93.sslip.io`

ENDPOINT		WHAT IT DOES	USES
Agenda and email			
POST	/summarize	One-line summary, to-do and urgency for each Canvas/calendar item. Already-seen items come from the cache, so only new ones cost a Gemini call.	Gemini, Tiger
GET	/mail	Recent inbox as one-liners, plus any deadline or meeting an email states.	Gmail, Gemini, Tiger
GET	/briefing	Today's plan, urgent email and training, spoken.	Gemini, ElevenLabs
Planning			
POST	/ripple	"My lab moved to Friday": what it collides with, free slots, cost against the budget. Overlaps are computed by plain code, not the model.	Gemini, Tiger
POST	/capture	Reads a flyer photo; checks your calendar, your interests and that week's budget.	Gemini, Backboard, Tiger
Voice, recall and Ask			
POST	/log	A voice note becomes a structured log (sets, reps, mood).	Gemini, Tiger, Backboard
GET	/recall	What you did on this weekday last time, with a suggestion.	Tiger, Backboard, Gemini
POST	/chat	Ask. Context: agenda, logs, email, budget, 120 days of history, memories.	Gemini, Tiger, Backboard
POST	/speak	Reads a reply aloud (the Listen button).	ElevenLabs
Memory and history			
GET	/memories	Everything remembered about you.	Backboard
PUT	/memories/{id}	Correct a memory.	Backboard
DELETE	/memories/{id}	Forget a memory.	Backboard
POST	/history	Record something that happened (habit done, work turned in). Re-sends are ignored.	Tiger
DELETE	/history/{ref}	Undo a record (habit unticked).	Tiger
Money			
GET	/money	This week's budget, spent, left, by category and entries.	Tiger
GET	/money/trend	Last 8 weeks by category (Tiger's time_bucket).	Tiger
POST	/spend	Log spending from a sentence or receipt photo. No amount: asks "How much was it?"	Gemini, Tiger

DELETE	/spend/{id}	Delete an entry.	Tiger
PUT	/budget	Set or clear the weekly budget.	Tiger
Status			
GET	/health	Is the server up and the database reachable.	Tiger

What happens in the background

WHEN YOU OPEN THE APP

- Pulls Canvas, iPhone Calendar and email; merges your classes, shifts and bills.
- Gets summaries for new items (`/summarize`) and this week's money.
- Rebuilds deadline, event, briefing and deadline-watch notifications.
- Refreshes habit reminders; removes reminders for classes past their end date.
- Records Canvas-reported submissions once.
- Updates the home-screen widget.

ABOUT EVERY 2 HOURS

- iOS may wake the app in the background to do the same refresh, so alerts stay current without opening it. iOS decides the exact timing.

WHEN YOU DO SOMETHING

- Tick a habit: saved on the phone, today's reminder cancelled, tick recorded in history.
- Mark work Turned In: recorded in history with the time.
- Ask something: your message is also sent to Backboard so lasting facts are remembered.
- Dictate: the mic stops by itself 1.6 s after you pause, after 8 s of silence, after 60 s, or when you leave the screen.

ON THE SERVER

- Summaries are cached in Tiger, so the same item is never sent to Gemini twice.
- If a Gemini model is busy or out of quota, the next one takes over.
- Dates and amounts are only read, never guessed; missing ones are asked for.

Notifications are scheduled on the phone, so reminders fire even with no internet. iOS allows 64 pending at once; habit reminders get first claim and the agenda uses what's left.

Where your data lives

PLACE	WHAT'S THERE
Your iPhone	Habits and ticks, classes, shifts and bills, settings, the Canvas link, scheduled notifications. Dictation audio stays here.
Tiger Data	Item summaries, email excerpts, voice logs, spending, the weekly budget, history.
Backboard	Memories about you, which you can see, correct or delete.
The server	The API keys and the Gmail sign-in token. None of them are in the app or the public repo.